

Aryan Choudhary

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Education

B.Tech – Industrial Internet of Things (IIoT)

Guru Gobind Singh Indraprastha University (IPU), Delhi

Expected Graduation: 2028

Relevant Coursework: Data Structures, Algorithms, Discrete Mathematics

Technical Skills

Programming Languages: C++, JavaScript

Web Technologies: HTML, CSS

Core CS: Data Structures, Algorithms, OOP, Operating Systems, Computer Networks

Databases: SQL (Basic)

Tools: Git, GitHub, VS Code

Projects

Smart Route Planner | C++

- Implemented Dijkstra's and A* algorithms to compute shortest and cost-efficient paths on weighted graphs.
- Optimized time complexity from $O(V^2)$ to $O((V+E) \log V)$ using priority queues (min-heaps).
- Modeled real-world constraints including traffic and toll costs using dynamic edge weights.
- Implemented path reconstruction and handled edge cases such as disconnected graphs and cycles.
- Benchmarked Dijkstra vs A* and analyzed time-space trade-offs.

Real-Time Collaborative Code Editor | JavaScript, Node.js

- Built a multi-user real-time code editor supporting concurrent users using WebSockets (Socket.IO).
- Designed an event-driven backend ensuring low-latency (<200ms) and consistent shared state.
- Implemented JWT-based authentication and room-level access control.
- Developed RESTful APIs using Node.js and Express for session and code management.
- Improved scalability through optimized data models and backend performance tuning.

Problem Solving

- Solved Data Structures and Algorithms problems in C++ on LeetCode.
- Strong hands-on practice in arrays, strings, recursion, trees, graphs, hashing, and dynamic programming.

Interests

Software Engineering, Algorithms, Artificial Intelligence